

EXP NO: 8 DATE:	TESTING – TEST PLANS AND TEST CASES
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AIM:-

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

Test Case Design Procedure

1. Understand Core Features of the Application

- User Signup & Login
- Student Enrollment in Courses
- Assignment Submission & Evaluation
- Viewing Grades and Attendance
- Admin Role Management and Report Generation

2. Define User Interactions

- Each test case simulates a real user behavior (e.g., logging in, submitting an assignment, viewing results).

3. Design Happy Path Test Cases

- Focused on validating that all core functionalities work correctly under normal conditions.
- **Example:** Student registers and logs in, enrolls in a course, submits assignment, and views grades.

4. Design Error Path Test Cases

- Simulate invalid inputs, system issues, or failed actions to ensure proper error handling.
- **Example:** Login with wrong credentials, submission without attachment, unauthorized access to admin panel.

5. Break Down Steps and Expected Results

- Each test case includes a clear sequence of actions and expected results.
- Ensures both manual testers and automation tools can follow the process easily.

6. Use Clear Naming and IDs

- Test cases are uniquely identified (e.g., TC01 – Student Login Success, TC12 – Invalid Assignment Submission).
- Facilitates easy mapping to features and tracking in Azure DevOps.

7. Separate Test Suites

- Grouped by functionality such as:
 - Login and Registration

- Course Enrollment
- Assignment Submission
- Report Generation
- Admin Functions
- Improves organization and enables focused execution in Azure DevOps.

8. Prioritize and Review

- High-priority assigned to critical workflows like login, course access, and grading.
- Reviewed for completeness, accuracy, and alignment with user stories and feature definitions.

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Student Management System– Test Plans

USER STORIES:-

- As a student, I want to sign up and log in securely so I can access academic services. (ID: 101)
- As a student, I need to enroll in courses to participate in classes. (ID: 102)
- As a student, I should be able to submit assignments online for evaluation. (ID: 103)
- As a student, I want to view grades and attendance in one place. (ID: 104)
- As an admin, I want to manage users and generate performance reports. (ID: 105)

Test Suites:-

Test suit : TS01 – User Login & Registration (ID: 201)

1. TC01 – Successful Student Sign-Up

- **Action:**
 - Go to the Sign-Up page
 - Enter valid name, email, and password
 - Click "Sign Up"
- **Expected Result:**
 - Student account is created and redirected to dashboard
- **Type:** Happy Path

2. TC02 – Login with Valid Credentials

- **Action:**
 - Go to Login page
 - Enter valid student email & password
 - Click "Login"
- **Expected Result:**
 - Logged in and taken to student dashboard
- **Type:** Happy Path

3. TC03 – Sign-Up with Existing Email

- **Action:**

- Go to Sign-Up
- Enter already registered email
- Click "Sign Up"

· **Expected Result:**

- Error message: "Email already registered"

· **Type:** Error Path

4. TC04 – Login with Incorrect Password

· **Action:**

- Enter wrong password
- Click "Login"

· **Expected Result:**

- Error: "Invalid username or password"

· **Type:** Error Path

Test Suite: TS02 – Course Enrollment (ID: 202)

1. TC05 – Enroll in Available Course

· **Action:**

- Login → Navigate to Course List
- Select a course → Click "Enroll"

· **Expected Result:**

- Enrollment confirmation shown

· **Type:** Happy Path

2. TC06 – Try Enrolling in a Full Course

· **Action:**

- Select a course with no seats available
- Click "Enroll"

· **Expected Result:**

- Error: "Enrollment failed – course full"

· **Type:** Error Path

Test Suite: TS03 – Assignment Submission (ID: 203)

1. TC07 – Submit Assignment Successfully

· **Action:**

- Login → Go to Assignments
- Upload file and click "Submit"

· **Expected Result:**

- Submission confirmation message shown

· **Type:** Happy Path

2. TC08 – Submit Without Uploading File

· **Action:**

- Open assignment submission

- Click "Submit" without file

· **Expected Result:**

- Error: "Please upload a file before

submitting" · **Type:** Error Path

Test Suite: TS04 – View Grades & Attendance (ID: 204)

1. TC09 – View Grades Dashboard

· **Action:**

- Login → Go to "My Grades"

· **Expected Result:**

- All grades are displayed by subject

· **Type:** Happy Path

2. TC10 – Fail to Fetch Grades

(Offline Scenario) · Action:

- Disconnect network → Go to "My Grades"

· **Expected Result:**

- Error: "Unable to fetch data – check

connection" · **Type:** Error Path

Test Suite: TS05 – Admin Management (ID: 205)

1. TC11 – Assign Role to New User

· **Action:**

- Login as Admin → Navigate to User Management
- Select user and assign role

· **Expected Result:**

- Confirmation: "Role assigned successfully"

· **Type:** Happy Path

2. TC12 – Generate Report with No Data

· **Action:**

- Navigate to Reports → Choose empty semester
- Click "Generate"

· **Expected Result:**

- Message: "No data available to

generate report" · **Type:** Error Path

3. Test Cases

4. Installation of test

5. Running the test cases
6. Recording the test case
7. Creating the bug

Test Results

The screenshot shows the 'Attendance management (ID: 17)' test results in Azure DevOps. The interface includes a 'Define' tab and an 'Execute' tab. Under the 'Execute' tab, there are 'Test Points (3 items)' listed in a table:

Title	Outcome
TC-01 Login with valid credentials	Failed
TC-02 Login with invalid credentials	Passed
TC-03 Login with empty username	Active

On the left, a 'Runner - Test Plans - Personal - Mic...' window shows the test steps for '18* TC-01 Login with valid credentials':

- Open login page (Status: Passed)
- Enter valid username (Status: Passed)
- Enter valid password (Status: Failed)
COMMENT: invalid password entered
- Click Login (Status: Failed)
COMMENT: Prev step failed

8. Test report summary

The screenshot shows a detailed view of a bug in Azure DevOps. The bug is titled '21 BG-01 Invalid password entered' and is assigned to 'Navdeep REC'. The bug is in the 'Active' state and is located in the 'STUDENT DATABASE MANAGEMENT' area, specifically in 'STUDENT DATABASE MANAGEMENT\Sprint 1'.

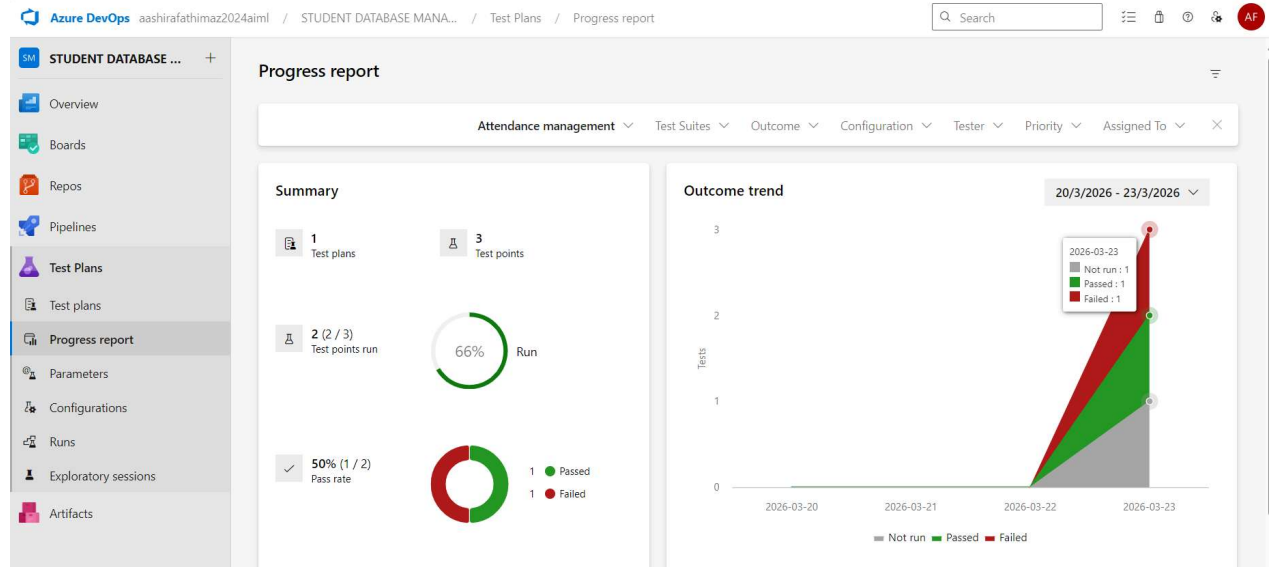
The 'Repro Steps' section shows the following steps:

Step no.	Result	Title
1.	Passed	Open login page
		Expected Result: User is redirected to dashboard
2.	Passed	Enter valid username
3.	None	Enter valid password
4.	None	Click Login

The 'Planning' section shows the bug's details, including 'Resolved Reason', 'Story Points', 'Priority', 'Severity', and 'Effort (Hours)'. The 'Deployment' section provides instructions on how to track releases associated with this work item.

1. Assigning bug to the developer and changing state

9. Progress report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path